68746: Advanced Quantum Science

Computer Lab Problems

Week 7 — Driven Qubits

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Methods

1. **Quantum evolution simulation methods:** Given a full-system Hamiltonian, consider a first-order Taylor expansion of the unitary gives:

$$U(t) = \exp(-itH/\hbar) \approx 1 - itH/\hbar \tag{1}$$

Consider the following two methods for simulating the dynamics of the system:

(a) At each time t, calculate the discrete-time unitary

$$\Delta U_n = \exp(-i\Delta t H\left[(t_n + t_{n-1})/2\right]/\hbar), \tag{2}$$

where $t_n = n\Delta t$, and update the quantum state according to

$$|\psi(t_n)\rangle = \Delta U_n |\psi(t_{n-1})\rangle. \tag{3}$$

(b) At each time t, calculate the quantum state update from the Hamiltonian at the mid-point of the time step

$$|\Delta\psi_n\rangle \approx -\frac{i\Delta t}{\hbar} H\left[(t_n + t_{n-1})/2\right]|\psi_{n-1}\rangle, \tag{4}$$

and update the quantum state according to

$$|\psi_n\rangle = |\psi_{n-1}\rangle - |\Delta\psi_n\rangle \tag{5}$$

Discuss the advantages and disadvantages of the two simulation methods.

Single-qubit rotations

2. **Qubit decoherence:**

(a) Write a python script to calculate the state evolution of a qubit under single-qubit X/Y rotations and plot the evolution on a Bloch sphere (e.g., using QuTiP).

(b) Now, in each step, use a uniformly distributed random sample to decide whether to apply a random bit flip or phase flip with probability p. For each random quantum evolution, calculate the Bloch coordinates. Average these coordinates over the random samples and plot the new evolutions on the Bloch sphere.

3. **Frequency noise:** Consider a simple Ramsey experiment defined by the following sequence: 1) a $X_{\pi/2}$ pulse, 2) variable time delay Δt, and then 3) a second $X_{\pi/2}$ pulse.

(a) Calculate the excited-state probability versus delay time.

(b) Assume the qubit experiences shot-to-shot frequency noise which you can model by adding a Gaussian noise to the free qubit frequency ω_0. Calculate the excited-state probability versus delay time for a random ensemble of shots and use that to calculate the average excited-state probability versus delay time.

(c) Now assume the qubit randomly swaps between two frequencies (with probabilities p_1 and p_2). Calculate the new average excited-state probability versus delay time.

Now modify the pulse sequence to add an additional X_π pulse in the middle of the delay time $\Delta t/2$.

(d) Recalculate the excited-state probability versus delay time for different random shots.

4. **Trotterised single-qubit evolution:** Consider the following single driven-qubit Hamiltonian:

$$H = \hbar\Omega_x\sigma_{\mathrm{x}} + \hbar\Omega_y\sigma_{\mathrm{y}}. \tag{6}$$

(a) Calculate the exact unitary qubit evolution for some given initial states.

(b) Calculate the approximate qubit evolution via Trotterisation for the same initial states, breaking up the evolution into small σ_{x} and σ_{y} steps, with Trotter step size τ_{step}. Plot the evolution for a range of different step sizes and compare against the exact evolution using state fidelity.

(c) Consider how to improve the evolution accuracy by symmetrising the Trotter step. Calculate the new evolution and compare for different Trotter step sizes.